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SUPERSEDING
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FEDERAL SPECIFICATION

LADDERS, SAFETY STEP (METAL, ROLLABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers safety metal stepladders mounted on casters. The casters are retractable by a lever arrangement or by the application of the weight of a person on the ladder.

1.2 Classification.

1.2.1 Types, classes, and sizes. Ladders shall be of the following types, classes, and sizes as specified (see 6.2).

Type I - Steel.
Type II - Aluminum.

Class 1 - Wide base - 25-1/2 to 36-1/2 inches wide.
Sizes - 1 to 12 steps inclusive, in one step increments.

Class 2 - Narrow base - 17-1/2 to 24 inches wide.
Sizes - 1 to 7 steps inclusive, in one step increments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Type I - Steel ladders shall be of plain carbon or alloyed steel at the manufacturers option. Type II aluminum ladders shall be of the 6000 series aluminum alloy in the T4, T5, or T6 temper at the manufacturer's option.

3.1.1 Casters. Casters shall be natural rubber (NR) or synthetic rubber (SBR) or any suitable molded plastic.

3.1.2 Tips for ladder legs. The tips for the ladder legs shall be made from natural rubber (NR) or synthetic rubber (SBR) or any suitable molded plastic.

3.2 Load rating. The design working load of ladder shall be on the basis of a 200 pound person with 50 pounds of equipment. The ladder shall be capable of supporting at least four times the design working load.

3.3 Ladder frame construction. Unless otherwise specified (see 6.2) both classes 1 and 2 of either Type I or II shall be completely or partially assembled at the option of the manufacturer. When partially assembled ladders are furnished, they shall consist of no more than 5 (five) sub-assemblies, sufficient bolts and self-locking nuts for assembly and a clear set of instructions. All ladders shall be capable of passing the tests specified in 4.4.

3.3.1 Pipe and tube. Steel and aluminum pipe and tube shall be welded, seamless extruded or drawn.

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3.4 Height and base depth. The height (working level) and base depth of ladders shall be in accordance with table I.

TABLE I. Height & depth of ladders

Size (Steps)	Height to top step (inches)	Depth or length at base (inches) minimum
1	10-12	13
2	18-20	19
3	27-30	25
4	36-40	30
5	45-50	36
6	54-60	42
7	65-70	47
8	75-80	55
9	84-90	60
10	94-100	66
11	103-110	72
12	113-120	78

3.4.1 Base width. Ladder frames shall have base width in accordance with table II.

TABLE II. Base widths

Class	Step increments	Base width inches1/
1	1 to 5	25-1/2 through 29-1/2
1	6 to 12	30 through 36-1/2
2	1 to 7	17-1/2 through 24

1/ Outside to outside dimension at bottom of side rails.

3.5 Steps and step slope. Steps shall be uniformly spaced and sloped, with a rise of not less than 9 inches, nor more than 10 inches, and a depth of not less than 7 inches and the top step shall not be less than 8 inches in depth. The slope of the steps section shall be a minimum of 55 degrees and a maximum of 60 degrees measured from the horizontal.

3.6 Slip resistant steps. The steps of the ladder shall have slip-resistant treads.

3.7 Handrails. Ladders having more than 5 steps or 60 inches verticle height to the top step shall be equipped with a handrail. The hand rails shall be a minimum of 29 inches high. Measurement shall be taken vertically from the certer of the step.

3.8 Casters. Casters shall have a bolt, rivet or bushing axle and swivel. Casters shall be furnished with ball bearing race on the swivel. The swivel type casters shall be spring loaded. The casters shall be properly designed for strength and dimension to support four times the design working load. The ladder stand shall have at least two locking casters or other means of locking the unit in position and at least two of the four casters shall be of the swivel type. Casters shall be in accordance with table III.

TABLE III. Minimum caster size, wheel diameter, inches

Class	Ladder size	Caster size
1	1-5	2
1	6-7	3
1	8-12	3 or 4
2	1-6	1-5/8 or 2
2	7	3

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3.9 Tips for ladder legs. Slip resistant, replaceable tips 3.2.2 shall be part of the bottom of all ladder legs to prevent slipping after the casters are disengaged.

3.10 Finish. Steel ladders shall be painted with aluminum or safety yellow paint as specified (see 6.2). Aluminum ladders may be uncoated, anodized or painted with safety yellow paint, as specified (see 6.2).

3.11 Performance.

3.11.1 Stability. When loaded and tested in accordance with 4.4.2 all legs shall be in contact with the ground or floor and ladders shall remain upright and all legs in contact with the testing surface when the horizontal pull is applied.

3.11.2 Compression strength. Ladders shall not exhibit any permanent deformation or structual failure when tested in accordance with 4.4.3.

3.11.3 Rigidity. The verticle deflection of the ladder when tested in accordance with 4.4.4.2.1 shall not exceed one-quarter inch. The difference between the deflection obtained in accordance with 4.4.4.2.1 shall not exceed one-quarter inch.

3.12 Identification and safety marking all ladders. Each ladder shall be legibly marked by stenciling, stamping or an affixed name plate with the size and the manufacturer's name, symbol or trademark and maximum safe load limit in pounds and kilograms. The marking shall also include a statement that the ladder is not to be used around electrical circuits or equipment.

3.13 Workmanship. All exposed metal surfaces shall be free from sharp edges, burrs, or other safety hazards. There shall be no structural defects including deformed steps, handrails, frames, or frame and handrail sections. Rivets in aluminum ladders shall be tightly driven. Welds in steel and aluminum ladders shall be free of cracks and closely spaced in-line surface porosity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Dimensional and visual examination. Sampling for these characteristics shall be in accordance with MIL-STD-105 inspection level S-4, acceptable quality level (AQL) 2.5 percent defective.

4.2.2 Sampling for tests. Ladders shall be sampled in accordance with MIL-STD-105 inspection level S-1. Failure of any sample to pass any of the tests shall be cause for rejection of the lot.

4.3 Examination and tests.

4.3.1 Dimensional and visual examination. Samples selected in accordance with 4.2.1 shall be examined for dimensional and visual defects in accordance with table IV.

TABLE IV. Classification of defects, end item

a.	Type, class and size not as specified
b.	Height or base depth and width not in accordance with tables I and II
c.	Handrails height not as specified 3.7
d.	Step depth less than required as specified in 3.5
e.	Top step depth less than required as specified in 3.5
f.	Steps not evenly spaced 3.5
g.	Tread on steps not in accordance with 3.6
h.	Casters not the correct type and size in accordance with table III
i.	Casters with locking device missing 3.8
j.	Painted metal surfaces of ladders not completely covered 3.10
k.	Workmanship not according to 3.13

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4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine whether packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table V. For examination of contents, the sample unit shall be on shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4 defects per hundred units.

TABLE V. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior)	Omitted; incorrect; illegible, or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.
Contents	Gross weight exceeds requirements of the applicable box specification.

4.3.3 Tests. Samples selected in accordance with 4.2.2 shall be tested as specified in 4.4. Failure of any sample to pass any of the tests shall be cause for rejection of the lot.

4.4 Test methods.

4.4.1 Preparation for test. Assemble partially assembled ladders using fasteners that are readily removable, except for the rigidity test.

4.4.2 Stability. Conduct two tests, one with a 90 pound load on the bottom step and the other with the same load on the middle or on the next step higher. Attach a portable scale near the mid-point of the vertical height and pull in a horizontal direction, at an angle of 90°, away from the steps until the scale registers 30 pounds. Sliding or skidding shall not be considered cause for rejection (see 3.11.1).

4.4.3 Compression. Place a load a 1000 pounds at the middle of the top step distributed over an area 3-1/2 inches wide by the depth of the step for ladder sizes 1 to 12. Repeat with the same load on the middle or 1 step higher for all ladders size 3 and larger (see 3.11.2).

4.4.4 Rigidity.

4.4.4.1 Assembly of partially assembled ladders. Assemble all partialy assembled ladders completely in accordance with the set of instructions specified in 3.3, using the same type lock fasteners furnished with the assembly. Tighten the fasteners using a uniform amount of torque. The amount of torque required shall be that amount required to properly seat the joint without any visible deformation of the metal.

4.4.4.2 Procedure. Test all fully assembled or partially assembled aluminum and steel ladders in accordance with 4.4.4.2.1 through 4.4.4.2.3.

4.4.4.2.1 Deflection, number 1. Mount the ladder upright on the steel horses placed in the shape of an "L". Clamp both corners of the supported side to the horses. Leave the other side with one corner supported, but unclamped. Leave the remaining corner unsupported. Mark the location of the supported legs on the horses so that the same measuring reference point can be achieved for deflection number 2. Load the unsupported corner with a 50 pound tensile load for one-half minute. (Apply the load to the bottom horizontal siderail as close to the corner as possible.) Release the load and measure the height from the surface of the floor to the bottom of the unsupported corner. Load the unsupported corner with a 250 pound tensile load for a period of one minute. Release the load and repeat the measurement from the surface of the floor to the bottom of the unsupported corner (see 3.11.3).

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4.4.4.2.2 Vibration. Remove the ladder from the horses and set it upright on a level floor inside a wooden frame. Construct the wooden frame using 2-inch by 4-inch lumber set on edge. The frame shall be large enough to permit a minimum of 1-inch of clearance between the inner surfaces of the frame and the outside edge of the ladder frame. Attach an eccentric-loaded vibrator on one side of the frame at the midway location. Bolt the vibrator to two, one-half inch metal plates straddling the side rail. Use lock washers under each nut. Vibrate the ladder using a range of 40 to 45 Hz inclusive for a period of one hour using a model DV 51/M vibrator, or equal (see 6.3).

4.4.4.2.3 Repeat the procedures of 4.4.4.2.1 (without tightening any fasteners for the partially assembled ladders). The difference between the deflection (deflection No. 1) and the deflection obtained in 4.4.2.3 shall not exceed the requirements as specified in 3.11.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified in (see 6.2).

5.1.1 Level A. Each ladder shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during multiple shipments, handling and storage.

5.1.2 Commercial. Each ladder shall be packaged as specified in 5.1.1.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Each ladder, packaged as specified in 5.1.1, shall be packed in corrugated fiberboard boxes made from weather resistant fiberboard with a bursting test strength of not less than 275 pounds per square inch (size permitting). The box flaps shall be secured with water resistant adhesive. Alternatively wire bound, cleated plywood or nailed wood boxes, exterior grade, shall be acceptable shipping containers when lined with waterproofed barrier material. The barrier material shall be sealed at the edges with water resistant tape or adhesive.

5.2.2 Commercial. Each ladder, packaged as specified in 5.1.2, shall be packed in fiberboard boxes to insure safe delivery at destination to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking. Unless otherwise specified (see 6.2) marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use.

6.1.1 Class 1 ladders. These ladders are intended for industrial use including maintenance operations.

6.1.2 Class 2 ladders. These ladders are intended for interior use in libraries, file rooms and stock rooms.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, class and size (1.2.1).
(c) Whether a specific type of construction is required (3.3).
(d) Finish (3.10)
(e) Selection of levels of packaging and packing (see 5.1 and 5.2).
(f) Whether marking is required (see 5.3).

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6.3 Vibrators. The model DV51/M vibrator is available from the Martin Engineering Co., U.S. Route 34, Dept. TR, Neponset, IL 61345 or equal.

Preparing Activity

GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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